Arpan Kumar Moharana

B.Tech in Artificial Intelligence & Data Science

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Summary

AI & Data Science enthusiast, graduate in Computer Science (AI & Data Science) from MIT World Peace University, Pune, with hands-on experience in machine learning, deep learning, and generative AI. Passionate about building intelligent, data-driven systems — from RAG pipelines and predictive models to AI-integrated applications. Skilled in Python, TensorFlow, PyTorch, and LangChain, with additional exposure to XR development and software engineering. Actively seeking AI, ML, and Data Science roles to apply research and engineering skills to real-world problems.

Education

MIT World Peace University, Pune, Maharashtra 2023 – 2026
B.Tech in Computer Science (AI & Data Science)

AISSMS College of Engineering, Pune , Maharashtra 2020 – 2023
Diploma in Computer Science

Skills

Languages: Python, SQL, R, NoSQL **Databases:** MySQL, MongoDB, Cassandra

AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, Predictive Modeling, Data Analysis, AI Agents, REST APIs

Frameworks & Tools: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, LangChain, Hugging Face, Sentence Transformers, Ollama, Flask, Docker, Git, Linux, Unity, OpenCV

Experience

XR Developer Intern July 2025 – Jan 2026
FireBirdVR, Pune *Tools: Unity, Blender, Git*

- Built 3+ interactive XR applications using Unity and Blender, contributing to full-cycle development across 4 industry verticals: education, healthcare, retail, and real estate.
- Applied 3D mathematics and spatial problem-solving in agile sprints, reducing feature delivery time by streamlining implementation workflows.
- Collaborated with a cross-functional team of 5+ developers to deliver high-performance immersive experiences on schedule.

Software Developer Intern Jul 2022 – Aug 2022
Globeminds Technology Pvt. Ltd., Pune *Tools: Java, MySQL, JDBC*

- Managed and secured MySQL databases for 2 Java-based web applications, maintaining data integrity across all production environments.
- Executed complete CRUD operations via JDBC, standardizing database workflows and reducing manual data handling overhead by 30%.

Projects

Brain Tumor Segmentation using Hybrid U-Net, Graph Neural Network, and SAM Jan 2026 – Present

- Engineered a hybrid deep learning pipeline combining U-Net, Graph Neural Networks, and SAM for brain tumor segmentation on the Brain Tumor Segmentation MRI dataset across 4 imaging modalities. *(Submitted to Springer Nature – Under Review)*
- Achieved 0.85 Dice Score, outperforming MobileNet (0.78) and standard U-Net (0.83) by up to 8.97% in segmentation accuracy and recall.
- Built end-to-end training and evaluation pipeline using PyTorch, PyTorch Geometric, and OpenCV with multimodal preprocessing (FLAIR, T1, T1-contrast, T2).

Semantic RAG-Based Question Answering System

- Constructed a fully local retrieval-augmented generation system using LangChain, ChromaDB, and Hugging Face MiniLM embeddings for document-based question answering with zero external API dependency.
- Established end-to-end pipeline covering ingestion of 100+ document chunks, text splitting, embedding generation, and cosine similarity retrieval.
- Evaluated retrieval quality using Hit-at-K and Mean Reciprocal Rank metrics; applied top-K (K=3) retrieval, improving answer relevance across all test queries.

Certifications

- Generative AI in Action – IBM SkillsBuild
- National Academic Immersion Program: ML & Deep Learning – NIT Goa
- GitHub Foundations – Microsoft Learn
- Unity Junior Programmer – Unity Technologies